A B C A L C U L U S

CHAPTER 2

Differentiation: Definition & Basic Rules

AP-Style Practice Questions

EASY MEDIUM HARD

Topics **2.1 – 2.10** AB

Name: _____ Period: _____

PART I

Topics 2.1 – 2.10

Multiple Choice Questions

Show all supporting work on scratch paper. On the AP Exam, Section I is split into a no-calculator and a calculator-allowed part — each question below is labeled accordingly.

Q1 EASY 2.1 Average Rate of Change No Calculator

The average rate of change of $f(x) = x^2 + 2x$ on $[1, 3]$ is

(A) 4

(B) 6

(C) 8

(D) 10

Q2 EASY 2.1 Definition of Derivative No Calculator

$\lim_{h \to 0} \frac{(3+h)^2 - 9}{h}$ equals

(A) 0

(B) 3

(C) 6

(D) 9

Q3 EASY 2.2 Connecting f and f' No Calculator

If f is differentiable at $x = a$, which of the following must be true?

(A) f is continuous at $x = a$.

(B) f has a local extremum at $x = a$.

(C) $\lim_{x \to a} f(x) = 0$.

(D) $f''(a)$ exists.

Q4 MEDIUM 2.2 Differentiability No Calculator

Which of the following is NOT differentiable at $x = 0$?

(A) $f(x) = x^2$

(B) $f(x) = |x|$

(C) $f(x) = \sin x$

(D) $f(x) = x^3$

Q5 EASY 2.3 Power Rule No Calculator

If $f(x) = 5x^4 - 3x^2 + 7$, then $f'(x) =$

(A) $20x^3 - 6x$

(B) $20x^3 - 6x + 7$

(C) $5x^3 - 3x$

(D) $20x^3 - 3x^2$

Q6 EASY 2.4 Trig Derivatives No Calculator

$$\frac{d}{dx}[\sin x - \cos x] =$$

(A) $\cos x - \sin x$

(B) $\cos x + \sin x$

(C) $-\cos x - \sin x$

(D) $-\sin x - \cos x$

Q7 MEDIUM 2.5 Exp/Log Derivatives No Calculator

$$\frac{d}{dx}\left[\, e^x \ln x \,\right] =$$

(A) $\dfrac{e^x}{x}$

(B) $e^x \ln x + \dfrac{e^x}{x}$

(C) $e^x \ln x$

(D) $e^x + \ln x$

Q8 MEDIUM 2.6 Product Rule No Calculator

If $h(x) = x^2 \cos x$, then $h'(x) =$

(A) $2x \cos x - x^2 \sin x$

(B) $2x \cos x + x^2 \sin x$

(C) $-2x \sin x$

(D) $2x - \sin x$

Q9 MEDIUM 2.7 Quotient Rule No Calculator

If $g(x) = \dfrac{x}{x^2+1}$, then $g'(x) =$

(A) $\dfrac{1}{2x}$

(B) $\dfrac{1-x^2}{(x^2+1)^2}$

(C) $\dfrac{x^2+1-2x^2}{x^2+1}$

(D) $\dfrac{x^2-1}{(x^2+1)^2}$

Q10 MEDIUM 2.8 Tangent Lines No Calculator

The tangent line to $f(x) = x^3 - 2x$ at $x = 1$ has equation

(A) $y = x - 2$

(B) $y = x - 1$

(C) $y = x - 3$

(D) $y = -x + 1$

Q11 MEDIUM 2.6 / 2.7 Rule Selection No Calculator

Let $f(x) = \dfrac{x \sin x}{e^x}$. Which differentiation rules are required?

(A) Product only

(B) Quotient only

(C) Product and quotient

(D) None (power rule suffices)

Q12 HARD 2.7 Quotient Rule (Table) No Calculator

Selected values of f, g, f', g' at $x = 2$ are given.

x	f	g	f'	g'
2	3	4	5	-1

If $h(x) = \dfrac{f(x)}{g(x)}$, then $h'(2) =$

(A) $\dfrac{17}{16}$

(B) $\dfrac{23}{16}$

(C) $\dfrac{13}{4}$

(D) $\dfrac{19}{16}$

Q13 MEDIUM 2.8 Normal Line No Calculator

The normal line to $y = \sqrt{x}$ at $x = 4$ has slope

(A) 4

(B) -4

(C) $\frac{1}{4}$

(D) $-\frac{1}{4}$

Q14 HARD 2.9 Higher Order No Calculator

If $f(x) = \sin(2x)$, then $f^{(4)}(x) =$

(A) $16\sin(2x)$

(B) $-16\sin(2x)$

(C) $16\cos(2x)$

(D) $-8\sin(2x)$

Q15 MEDIUM 2.10 Motion No Calculator

A particle's position is $s(t) = t^3 - 6t^2 + 9t$. The acceleration at $t = 3$ is

(A) 0

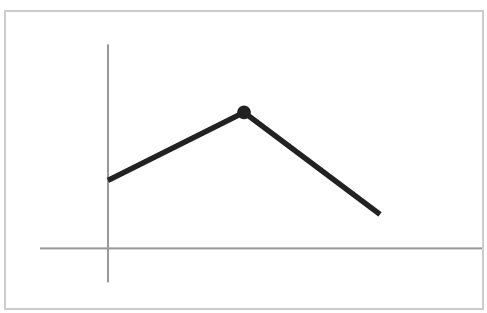
(B) 3

(C) 6

(D) 12

Q16 HARD 2.2 Graphical Reading No Calculator

The graph of f consists of two line segments meeting at a sharp corner at $x = 3$. Which statement about f' is true?



(A) $f'(3)$ exists and equals zero.

(B) f' has a jump at $x = 3$; $f'(3)$ does not exist.

(C) f is not continuous at $x = 3$.

(D) f' is constant on $(1, 5)$.

Q17 MEDIUM 2.5 a^x & $\log_a$ No Calculator

$$\frac{d}{dx}\left[\,3^x\,\right] =$$

(A) 3^x

(B) $x \cdot 3^{x-1}$

(C) $3^x \ln 3$

(D) $\dfrac{3^x}{\ln 3}$

Q18 HARD 2.8 Calculator Tangent Calculator

Let $f(x) = x\,e^{-x}$. The value of $x > 0$ at which the tangent line to $y = f(x)$ is horizontal is closest to

(A) 0.5

(B) 1.0

(C) 1.5

(D) 2.0

Q19 MEDIUM 2.1 Limit-Definition Disguise No Calculator

$$\lim_{h \to 0} \frac{\sin\left(\frac{\pi}{3} + h\right) - \sin\left(\frac{\pi}{3}\right)}{h} =$$

(A) 0

(B) $\dfrac{1}{2}$

(C) $\dfrac{\sqrt{3}}{2}$

(D) 1

Q20 MEDIUM 2.1 Limit-Definition Disguise (Exp) No Calculator

$$\lim_{h \to 0} \frac{e^{2+h} - e^2}{h} =$$

(A) 0

(B) 1

(C) e

(D) e^2

Q21 HARD 2.2 Cusp / Vertical Tangent No Calculator

Consider $f(x) = x^{2/3}$. Which statement about f at $x = 0$ is true?

(A) f is differentiable at $x = 0$ with $f'(0) = 0$.

(B) f has a vertical tangent at $x = 0$ — the one-sided limits of f' are both $+\infty$.

(C) f has a cusp at $x = 0$ — f is continuous there, but the one-sided limits of f' at 0 are infinite with opposite signs.

(D) f is not continuous at $x = 0$.

PART II *Show All Work*

Free-Response Questions

Free-response answers require complete setup: derivative rules stated, clear notation, and units for contextual problems. Tangent-line equations should be in point-slope or slope-intercept form.

FRQ 1 EASY 2.1 / 2.3 Limit Definition No Calculator

Let $f(x) = x^2 - 4x + 1$.

(a) Use the limit definition of the derivative to compute $f'(x)$.

(b) Find $f'(3)$ and interpret it as a rate of change at $x = 3$.

(c) Find the equation of the tangent line to $y = f(x)$ at $x = 3$.

FRQ 2 MEDIUM 2.6 / 2.7 Rules with Table No Calculator

The table gives values of f, g, f', g' at $x = 1$ and $x = 3$.

x	f	g	f'	g'
1	2	4	-1	3
3	5	-2	2	1

(a) Find $\dfrac{d}{dx}[\, f(x)g(x)\,]$ at $x = 1$.

(b) Find $\dfrac{d}{dx}\left[\dfrac{f(x)}{g(x)}\right]$ at $x = 3$.

(c) Let $h(x) = x \cdot g(x)$. Write the equation of the tangent line to $y = h(x)$ at $x = 3$.

FRQ 3 MEDIUM 2.10 Motion No Calculator

A particle moves along the x-axis with position $s(t) = t^3 - 9t^2 + 15t + 4$ for $t \geq 0$ (seconds, meters).

(a) Find the velocity $v(t)$ and acceleration $a(t)$.

(b) Find all times $t \geq 0$ when the particle is at rest.

(c) Determine the intervals on which the particle is moving right, and those on which it is moving left. Justify.

(d) At $t = 2$, is the particle speeding up or slowing down? Justify using the signs of v and a.

FRQ 4 HARD 2.2 Differentiability & Continuity No Calculator

Let $f(x) = \begin{cases} x^2 + ax, & x \le 1 \\ bx + c, & x > 1 \end{cases}$.

(a) State the condition for f to be continuous at $x = 1$ in terms of a, b, c.

(b) State the condition for f to be differentiable at $x = 1$ in terms of a, b.

(c) Given that $f(1) = 3$ and f is differentiable at $x = 1$, solve for a, b, and c.

(d) With the values from (c), write $f'(x)$ as a piecewise function and state $f'(1)$.

FRQ 5 HARD 2.8 Tangent & Parameters No Calculator

Let $f(x) = x^3 + px^2 + qx$.

(a) Find $f'(x)$ in terms of p, q.

(b) The graph of $y = f(x)$ passes through $(1, 2)$ with horizontal tangent at $x = 1$. Set up and solve a system for p and q.

(c) Using the values from (b), find the equation of the tangent line at $x = -1$.

(d) Find all x where the tangent line to $y = f(x)$ has slope 6.